

Effects of urban morphology on anthropogenic heat dispersion in tropical high-density residential areas

A.S. Adelia, C. Yuan*, L. Liu, R.Q. Shan

Department of Architecture, School of Design and Environment, National University of Singapore, Singapore

ARTICLE INFO

Article history:

Received 22 June 2018

Revised 16 December 2018

Accepted 3 January 2019

Available online 23 January 2019

Keywords:

Anthropogenic heat dispersion

Buoyancy effect

Building design

CFD

Urban heat island

ABSTRACT

Anthropogenic heat is constantly emitted to the urban environment in which could trigger a vicious cycle: risen the air temperature, subsequently increase the cooling-energy consumption and further intensify the urban heat island effect. While extensive studies had been conducted to estimate the magnitude of anthropogenic heat emission in urban areas, research that investigates the behaviour of anthropogenic heat dispersion in the outdoor environment is limited. This research aims to understand the interaction between the airflow and the flux of heat rejected from the high-rise residential buildings, and effects of heat flux on the air temperature in the street canyon. Parametric study is performed by using Computational Fluid Dynamics (CFD) simulation method, and the results indicate that, emission position and building porosity play an important role in anthropogenic heat removal from the street canyon. Single strategy, such as building height variance, can help to reduce the street canyon temperature to some extents, but the impact can be more significant when it is coupled with other strategies. At the end, a case study is conducted to evaluate the practicability of the suggested mitigation strategies in real urban design practice. This study bridges the knowledge gap between the complex aerodynamic physics and the real urban planning and design practices.

© 2019 Elsevier B.V. All rights reserved.

1. Introduction

Due to rapid urbanisation, high energy consumption for cooling in tropical cities is aggravated by the increasing urban population, causes larger anthropogenic emissions produced, such as heat. In urban areas, building sector is known as the largest energy consumer, accounting for nearly 40% of the total energy consumption hence making it as the largest contributor of global heat emission (e.g. [1,2]). Building air-conditioning (AC) system requires the largest portion of building electricity demand and has a direct effect on the urban thermal environment by rejecting the accumulated heat from the building's interior to the urban atmosphere in order to achieve indoor thermal comfort [3,4]. This rejected heat can spatially and temporally alter the near-surface temperatures, which in turn can lead to higher cooling energy demand, especially in tropical cities. While air temperature in the city blocks is highly influenced by its structure, geometry, land cover and metabolism (e.g. [5,6]), long and deep street canopies often found in megacities can block air flow to penetrate and thus trap the anthropogenic heat emitted from AC at the urban canyon layer. Stagnated airflow caused by low wind permeability in the urban canyon limits the

dispersion potential for the anthropogenic heat, which can further increase the ambient air temperature.

The above coupling effects (i.e. high anthropogenic heat generation from buildings and low dispersion potential) have led to frequent reports of high urban heat island (UHI) in metropolitan cities. Kimura and Takahashi [7] and Ichinose et al. [8] investigated the effect of land use and anthropogenic heating on the urban air temperature in Tokyo and reported that most of UHI at night is caused by anthropogenic heating and low wind speed. By simulating a residential area in Taipei, Hsieh et al. [9] observed that the air temperature around buildings tends to be higher due to rejected heat from cooling system and found out that as the air temperature rises, more energy is consumed for cooling processes. In Tokyo's commercial district, the magnitude of anthropogenic heat from air-conditioners is more than 1.5 times larger than the sensible heat released from the building surface during summer [10]. Due to the significance of the anthropogenic heat released from air-conditioning system, several studies have been undertaken to investigate the influence of anthropogenic heat from air-conditioners in urban thermal environment in Asian megacities, such as Tokyo, Hong Kong and Singapore (e.g. [3,11–13]). While many of these literatures were focused on the methodology of anthropogenic heat emission estimation and its impact on UHI effect, research that is investigating the anthropogenic heat disper-

* Corresponding author.

E-mail address: akiyuan@nus.edu.sg (C. Yuan).



Fig. 1. Real urban settings of Singapore's high-density residential estate.

sion potentials caused by different urban geometries is largely unexplored in urban modelling studies.

Anthropogenic heat release in the street canyon can increase the temperature especially the areas near the heat sources, where the influence of buoyancy can be found and affect the flow field around buildings. To study the thermal effects on flow and dispersion in street canyon, there are three potential modelling methods: field measurement (e.g. [14]), wind/water tunnel experiment (e.g. [15–17]) and computational fluid dynamics (CFD) simulation (e.g. [18–20]). These studies show that buoyancy effect is more apparent in high surface temperature and low wind speed condition. The effects can also change, depend on different heating position and intensity. Recently, Allegrini [21] conducted a wind tunnel experiment with ground heating and measured the three-dimensional flow structures of buoyant and forced convective flow in street canyon using particle image velocimetry (PIV). He concluded that lateral flow has an influence in the formation of buoyant flow structures, especially in the mixed convective case. Similar finding was reported in Kim and Baik [22]. However, in most of the aforementioned studies, buoyancy effect was usually induced by the increase in surface temperatures to reproduce the effect of solar radiation. It is different to the scenario in this paper that anthropogenic heat is rejected from AC as hot air with emission velocity.

The aim of this study is to clarify the mechanism of anthropogenic heat dispersion with regards to different urban scenarios. Singapore is chosen as our case study that represents high density tropical cities, where dense and compact living is inevitable due to limited land area available. This study focuses on residential areas, where the impact of AC anthropogenic heat is expected to be much significant as compared to other land uses in the street canyon level, due to the AC type and location of the AC outdoor unit on the residential building façade. In Singapore, high-density public housing (HDB) dominates the residential area across the island (Fig. 1). In 2016, majority of Singaporean lived in HDB (80%), whereas only 13% of Singaporean lived in Condominium [23]. Therefore, HDB is taken as the base model in the simulation and different design scenarios are developed in the parametric study. Scientific understandings and corresponding mitigation strategies are derived from the computational parametric study, which are applicable for urban design and planning practice. This would help designers and planners in their decision making, without the need to perform CFD simulation or complicated calculation associated with the fluid mechanics.

In this paper, computational parametric study is conducted to investigate the anthropogenic heat dispersion at high density residential areas during nighttime when the use of AC system is at the highest. First, using morphological data derived from geo-

graphic information system (GIS) and existing guidelines, the real urban settings for typical high-density residential area in Singapore is simplified in Section 2. Second, the modelling method is discussed in Section 3, including numerical dispersion model, anthropogenic heat estimation as input boundary condition, and turbulence model. Third, cases with various design strategies are developed in the parametric study and CFD simulations are conducted to evaluate the impact of different urban morphologies in terms of the distribution of heat emission and ambient air temperature in the street-canyon scale in Section 4. At last, proposed mitigation strategies that are derived from these new understandings are discussed for real urban planning and design application.

2. Modelling methods

The injection of heat emission in the outdoor urban environment causes three types of heat transfer: 1. Natural convection by buoyancy effect, 2. Forced convection by the external flow, and 3. Mixture of natural and forced convection [24]. The additional heat input creates the thermally driven buoyant flow allowing the cool air to lift the warm air up due to the force of gravity acting on the fluid density variations, also known as natural convection. Whereas, the forced convection is the type of heat transfer generated by the external sources (e.g. fan, pump, etc.), in this case is the outdoor airflow and the hot jets coming from the emission sources. The CFD commercial code ANSYS Fluent was used to simulate the mixed convection problem of anthropogenic heat release from air-conditioner systems. To model the natural convection in the flow field, Boussinesq model is applied in the simulation that uses the Boussinesq approximation to simplify basic equations. The basis of this approximation is when the variation of temperature in the domain is small, density differences also varies little and thus the buoyancy drives the motion of fluid. Therefore in this study, fluid density (ρ) is assumed to be constant everywhere except when it coupled with gravitational acceleration (g) in the body force term to represent the buoyancy force in the momentum equation [25]:

$$(\rho - \rho_0)g \approx \rho_0\beta(T - T_0)g \quad (1)$$

where ρ_0 is the (constant) reference density of the flow, β is the thermal expansion coefficient, T is fluid temperature, T_0 is the reference temperature, and g is the acceleration due to gravity force. It should be noted that this model is only valid if $\beta(T - T_0) \ll 1$.

2.1. Validation

In this study, the RANS SST $k-\omega$ model was validated by comparing its results with wind tunnel data from Allegrini et al. [17].

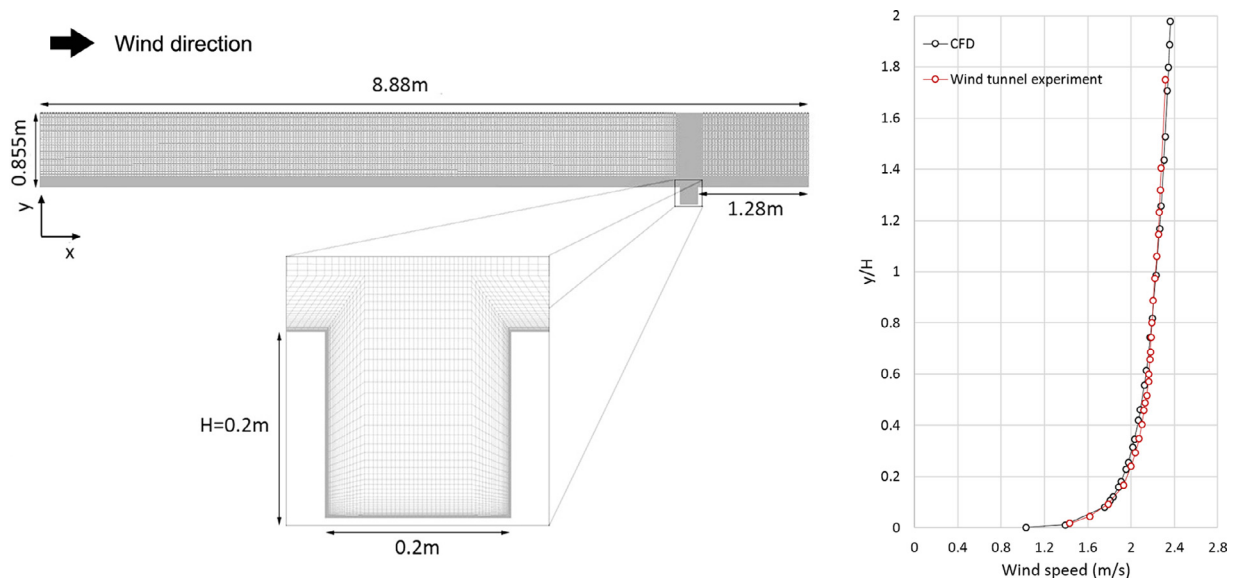


Fig. 2. Simulation settings [17]. Left: computational domain and grid structure; Right: inlet wind velocity profile.

All the simulation settings, i.e. domain configurations and boundary conditions, are the same as the ones at the wind tunnel experiment. Street canyon was modelled as the cavity, and the size of the domain and street canyon are 0.855×8.88 m and 0.2×0.2 m ($W \times L$) respectively, as shown in Fig. 2. The flow direction is normal to the street canyon, and the wind velocity profile at the inlet was set following the wind profile in wind tunnel experiment (Fig. 2). Symmetry and pressure outlet were imposed at top boundary and outlet respectively. The temperature of all three surfaces of the street canyon were set as 70°C to simulate the heated surfaces and incoming air temperature was 23°C . With 2.32 m/s reference velocity (U_{ref}), Froude number (Fr) of 17.29 was used in this validation study. Structural grid type was used for the mesh with 33,738 number of cells. As shown at Fig. 2, the mesh was refined near

all the three surfaces (i.e. wall and ground surface) of the street canyon with the grid ratio of 1.3, followed the Architectural Institute of Japan (AIJ) guidelines [26].

Fig. 3 shows the cross-comparison between simulation result using SST $k-\omega$ model and wind tunnel experiment. The normalised horizontal velocity was collected at the test line located at the centre of the street canyon. Linear regression analysis was performed to evaluate the correlation of CFD and wind tunnel data. The analysis indicates that there is a good agreement between SST $k-\omega$ and wind tunnel data ($R^2=0.97$), as shown in Fig. 3. This validation result suggests that SST $k-\omega$ can reproduce the mixed convection, caused by input air flow and buoyancy effect, within the street canyon. Therefore, SST $k-\omega$ model was used in the following parametric study.

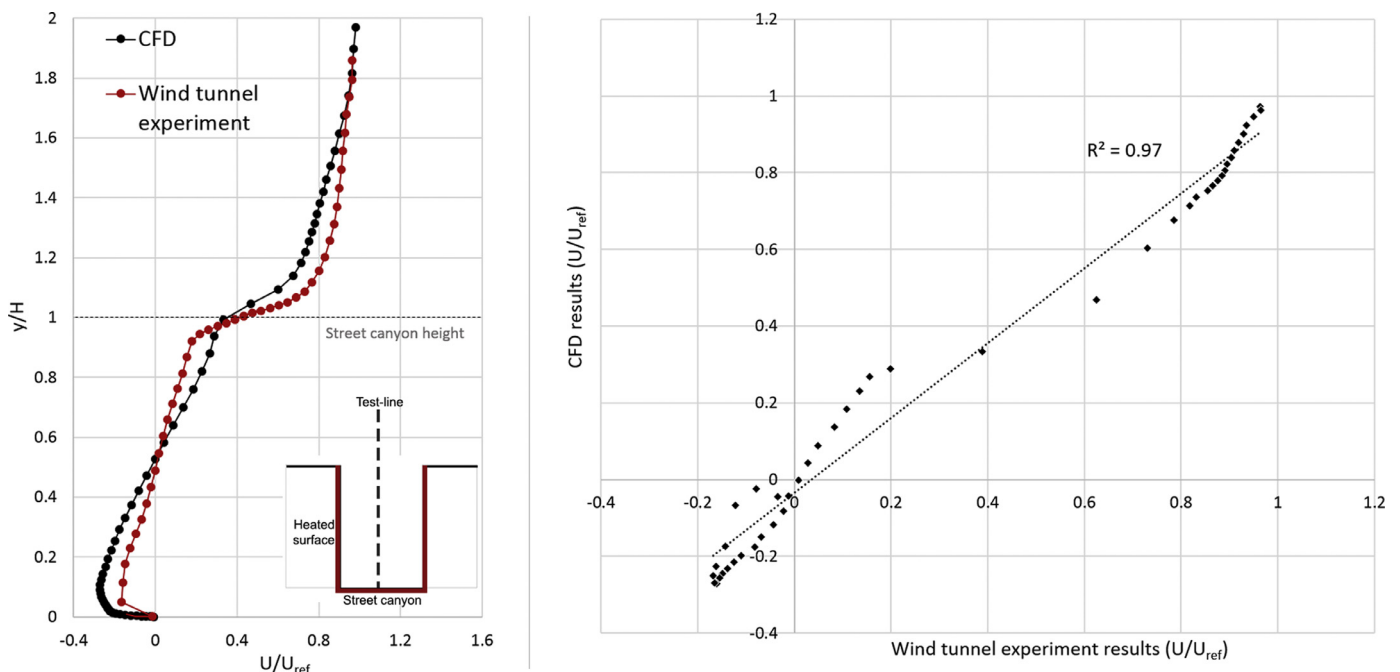


Fig. 3. Validation results. Left: Normalised horizontal velocity profile (U/U_{ref}) from CFD simulation and wind tunnel experiment; Right: Linear regression analysis result.



Fig. 4. AC outdoor unit configuration in HDB blocks. The heat sources are highlighted in red. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2.2. Input boundary conditions (Heat emission)

Different heating positions as well as heating intensity can significantly alter the thermal effect on the mean flow (e.g. [18,19]). In this study, the heat emission boundary conditions include: 1) heat source location, 2) anthropogenic heat from sources [i.e., the increase of ambient air temperature (ΔT)], and 3) emission velocity of exhaust air (V_{out}). In Singapore HDB, split-AC system is typically installed in the apartment unit. Such system uses condenser (outdoor unit) as heat exchanger device to dissipate the anthropogenic heat, produced by indoor activities, to the outside environment, and is normally attached on the exterior wall of the buildings. Since these condensers are normally arranged vertically on the exterior wall in real condition, we modelled them as a vertical strip instead of individual outlet to avoid high computational cost due to the modelling of individual AC condenser unit. Therefore, the heat source location in this study is shown in Fig. 4, highlighted as the red areas.

Anthropogenic heat from the split-AC operation of each apartment (Q_A) consists of cooling loads (Q_c) produced in the interior spaces, and cooling energy (E_c) needed for the AC operation to remove generated heat (Fig. 5). The anthropogenic heat by a condenser unit is described by Kikegawa et al. [27] as the following:

$$Q_A = Q_c + E_c, \text{ where } E_c = Q_c / \text{COP} \quad (2)$$

$$\text{Thus, } Q_A = Q_c \left(\frac{\text{COP} + 1}{\text{COP}} \right) \quad (3)$$

where Q_c is 2.5 KW per apartment, assuming only one split-AC unit operating in full-load capacity in the apartment. Q_c is derived based on the total air-conditioned area (i.e. bedroom). This calculation is provided by the Energy Efficient (E2) Singapore (<http://www.e2singapore.gov.sg>). The typical bedroom floor area of HDB flat was estimated based on HDB design guidelines and statistic report on 2016–2017 calendar years [28,29]. E_c is the quotient of

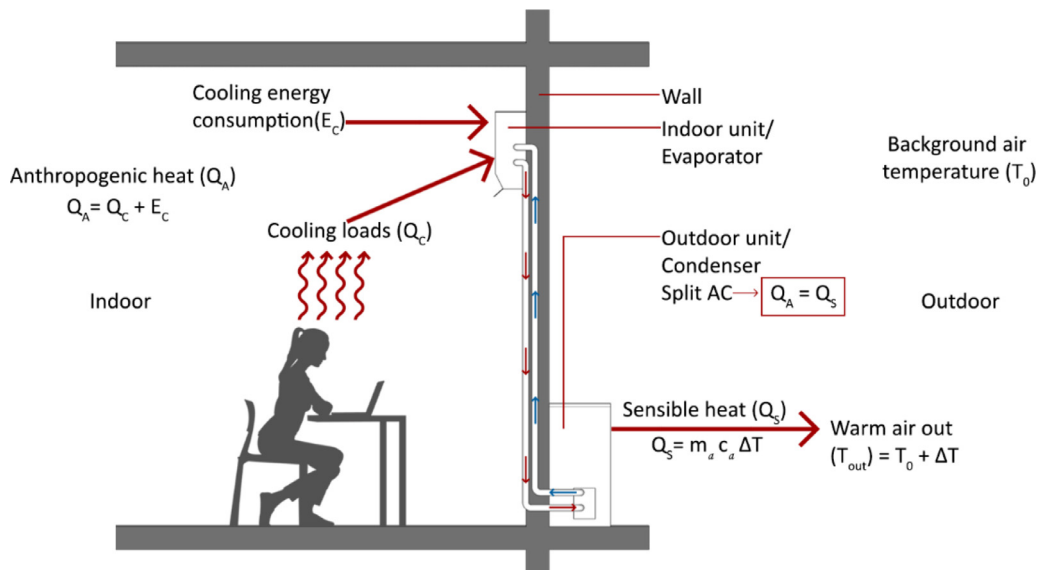


Fig. 5. Diagram of the estimation method of anthropogenic heat generated from air-conditioner unit. The energy balance between anthropogenic heat (Q_A) and sensible heat (Q_s) is highlighted.

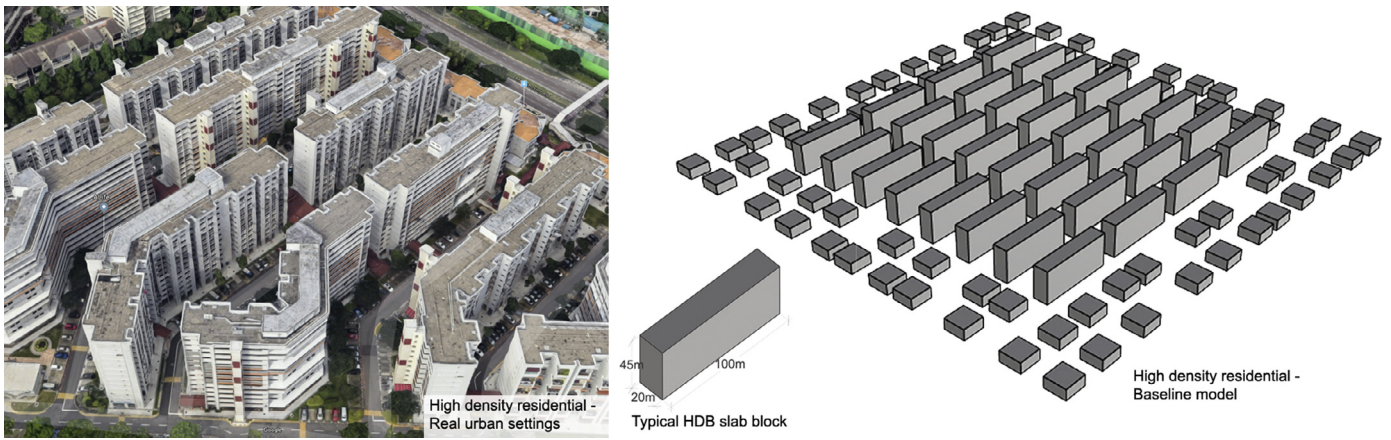


Fig. 6. Real urban settings for residential estate and its corresponding parametric model.

Q_c and the coefficient of performance (COP) for the AC operation (Eq. 2), in which COP equals to 3.34 that was adopted from energy label and tick rating of AC operation in the households of Singapore [30]. Therefore, the resultant Q_A turns out to be 3.25 KW per apartment unit.

We validated the above estimation by cross-comparing it with the Singapore anthropogenic heat estimation conducted by Quah and Roth [31]. They estimated $\bar{Q}$ (i.e., anthropogenic heat per m^2) based on the household electricity consumption data, and showed a 6–8 W/m^2 range of $\bar{Q}$ for the HDB category. In this study, we assumed that 50% of the total apartment units in one building block operate in full capacity to represent the normal household daily AC usage in Singapore. Thus, the total anthropogenic heat of an HDB block can be calculated by multiplying Q_A by the number of operating apartment units, and subsequently normalized by total floor area, yielding a $\bar{Q}$ of 7.6 W/m^2 , which is consistent with the estimation by Quah and Roth [31].

Anthropogenic heat flux (Q_A) was estimated based on energy balance, as shown in Fig. 5. Condenser unit as heat exchanger device releases anthropogenic heat from split-AC operation in a form of sensible heat (Q_S). Sensible heat increases the dry bulb temperature, whereas the humidity ratio remains the same due to no changes in moisture. This statement can be expressed as [3]:

$$Q_A = Q_S, \text{ where } Q_S = m_a c_a \Delta T \quad (4)$$

where m_a (air flow rate) is set to 245 kg/s [24], and c_a (specific heat of air) is set to 1.005 kJ/kg.K, resulting in a ΔT of 13 °C, which is in agreement with the 10–13 °C range of temperature increase near the split-type condenser unit in Singapore measured by [32]. The temperature of the exhaust air (T_{out}) is the addition of ΔT and the ambient air temperature (T_0), which is 27 °C, resulting in $T_{out} = 40$ °C, which is used in the following CFD simulation.

At last, we calculated the emission velocity of exhaust air (v) using mass flow rate formula as:

$$m_a = \rho v A \quad (5)$$

where ρ is the air density, and A is the area of heat source. As mentioned above, instead of modelling every single outlet for individual condenser unit, we grouped all condenser units located in one vertical stretch (i.e. the red strips shown in Fig. 4), and applied the spatial averaged emission velocity. Therefore, we used the area of red strip (1 m × 42 m), as area of heat source (A), to calculate v , which is 0.03 m/s.

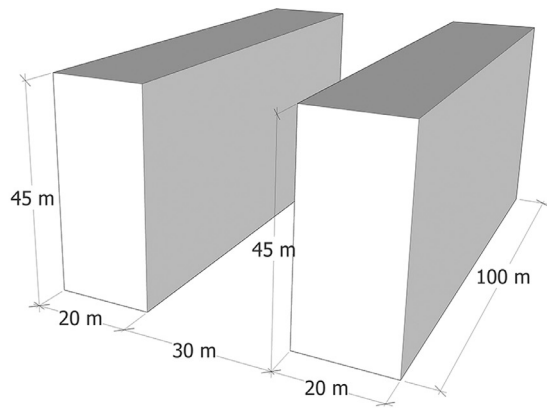
3. Computational parametric study

3.1. Parametric model

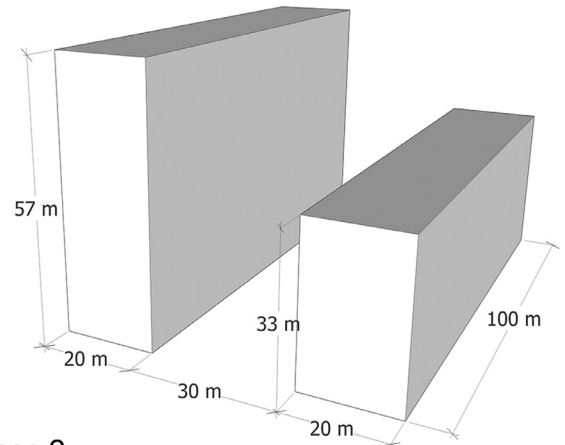
In this parametric study, the real urban morphology at residential areas was modelled in two stages at building and neighbourhood scales. The first stage is building scale model, where the most common typology of a single HDB block was simplified and parametrized based on existing guidelines and building codes [28,29]. The typical HDB block layout is a series of residential units along one side of a long corridor, so called the slab block typology. This layout was then simplified as a simple rectangular geometry. The model's building footprint is 2000 m^2 , derived from the average ground floor area computed in GIS, while the model's building height, 45 m (i.e. 15 storeys), was obtained from the maximum building height allowed for slab block HDB.

In the second stage at the neighbourhood scale, the typical HDB block was then replicated and arranged in larger scale that corresponds to the HDB's urban index, e.g. the minimum distance between two blocks, site coverage and floor area ratio (FAR). In HDB town planning, the residential land use ranges between 30–40% of the total HDB town area and FAR is from 2.8–4.2 [33]. According to the HDB guidelines [28,29], there is a minimum distance of 20 m between residential blocks. The number of blocks is controlled by the site coverage and FAR while the distance between blocks is the result of the block arrangement, to ensure that the density of the real urban settings and the parametric model match with each other. Using these requirements, individual building block was then arranged in a 500 × 500 m^2 study area, and a simplified neighbourhood scale HDB building blocks model was developed as shown in Fig. 6, similar to other previous studies [31,34,35]. This simplified model was applied as the baseline case in the parametric study to evaluate the performance of mitigation strategies.

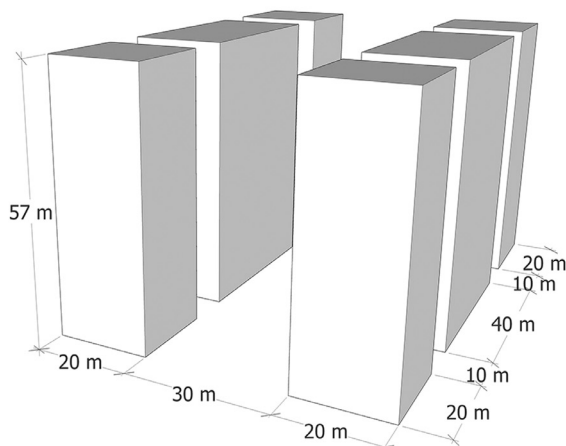
Based on the baseline case developed in this study, a total of twelve cases, i.e. six cases for each leeward/windward heat source location, were simulated in the parametric model to evaluate the relationship between different building geometry and anthropogenic heat dispersion, as shown in Fig. 7. Case 1 (i.e. Baseline model) refers to the current most typical HDB apartment blocks (slab block type), Case 2 introduces building height variance scenario that is known for a better urban flow distribution. Case 3 represents the building separation strategy where site coverage is reduced to create more wind corridors, whereas building voids are included in Case 4. While in Cases 2, 3 and 4, single modification is established for each case, multiple strategies are included in Cases 5 and 6. In these two cases, combined effect is expected, building height variance is coupled with building voids strategy in Case



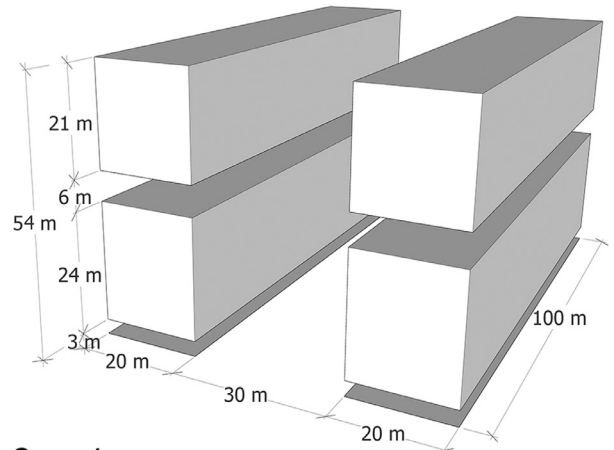
Case 1 (Baseline Case)



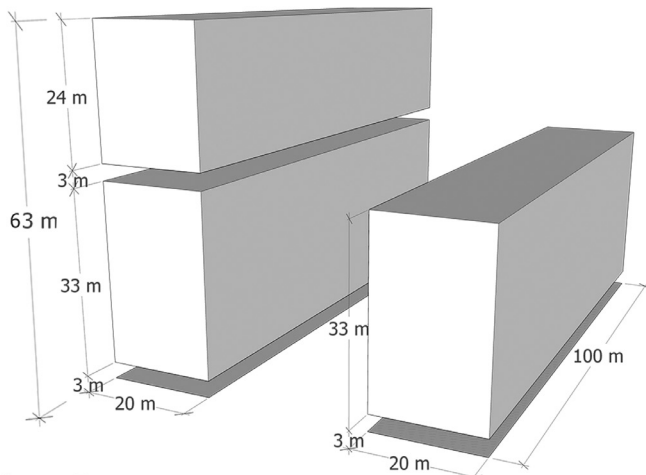
Case 2



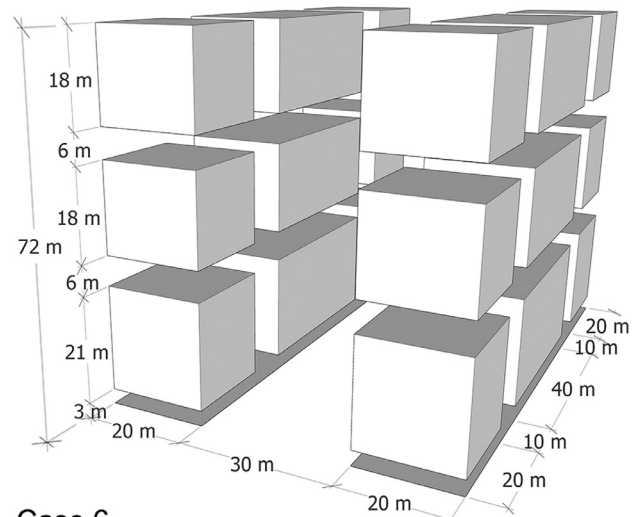
Case 3



Case 4



Case 5



Case 6

Fig. 7. Six cases in the parametric study. Case 1 (Baseline case) is simplified from real urban scenario, and cases 2–6 represent various mitigation design strategies.

5, while the latter is coupled with building separation strategy in Case 6. It should be noted that urban density (i.e. Floor Area Ratio (FAR)) of 4.8 is remained constant across all cases in order to ensure that all these cases are comparable. Detailed information for all cases is summarised in [Table 1](#).

Besides building typology parameters, the location of emission sources is considered as a design parameter, i.e. leeward emission and windward emission. It is not only because the position of AC condenser is an important design decision, but also because the

Table 1

Six cases representing 12 scenarios.

Cases	Design strategy		Heat emission location	Building height 1/Building height 2(m)	FAR
	Type	Parameters			
Case 1*	None	Existing urban form (Slab block)	Leeward / Windward	45/45	4.8
Case 2	Single	Building height variance		57/33	
Case 3	Single	Building separation (Point block)		57/57	
Case 4	Single	Building void		54/54	
Case 5	Coupled	Building height variance + building void		63/33	
Case 6	Coupled	Point block + building void		72/72	

* Baseline Case

relative position to the incoming wind direction is significant to heat dispersion.

3.2. Simulation settings

The simulation configuration is shown in Fig. 8, which follows the AIJ guideline [26]. The domain size is 1500 m × 1500 m × 300 m, in which the grid numbers are from 7 million to 17 million, different case by case. The target area is located in the centre, and the surround building blocks are arranged to represent the urban context as we did in the previous studies [36,37]. CFD settings are the same with the ones applied in the validation study, as shown in Table 2. The input wind direction is normal to the street canyon. Wind profile power law equation was applied as the inlet boundary condition, where α is 0.27, and wind speed of 10 m/s at the 300 m reference height based on the annual-averaged wind infor-

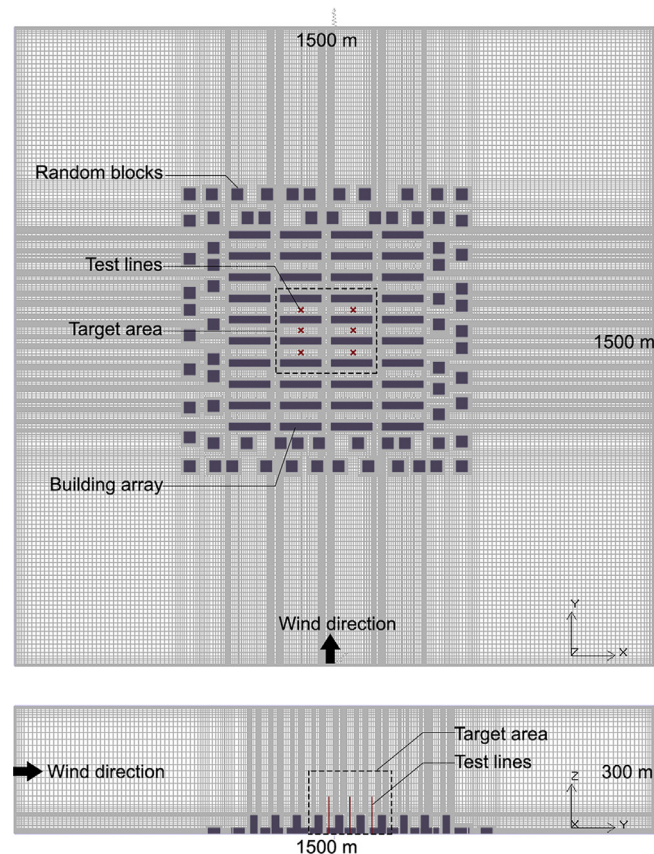


Fig. 8. Modelling configurations of the parametric study. Test lines have been highlighted in red. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2

CFD simulation settings.

Turbulence model	SST $k-\omega$
Computational grid type	Structural grid
Blockage ratio	< 5%
Grid expansion ratio	1.3
Grid resolution	Grid size less than 1/10 building length
Density	Boussinesq
Input wind profile	power law equation
Operating temperature	27 °C
Heat emission temperature	40 °C
Emission velocity	0.03 m/s
Convergence criteria	Below 1E-5

mation collected at Changi airport station.

$$U_{(h)} = U_{met} \left(\frac{h}{d_{met}} \right)^{\alpha} \quad (6)$$

where U_{met} is the mean wind speed at the reference height, d_{met} is the reference height, α surface roughness factor.

In the CFD simulation, heat emission sources are introduced as velocity inlet, and placed on either side of the street canyon (i.e. leeward or windward wall). The temperature and velocity input of the hot jet expelled from the heat sources is derived from the calculation described in the Section 2.2. Here, mixed convective flow is simulated using Froude number (Fr) 0.78, that is defined as [16]:

$$Fr = \frac{U_{ref}^2}{gh \frac{T_A - T_{ref}}{T_{ref}}} \quad (7)$$

where U_{ref} is reference velocity, h is reference height, T_A is the temperature of anthropogenic heat emission discharge from building, T_{ref} is ambient temperature and g is gravitational acceleration.

4. Modelling results and discussion

The modelling results were summarized in Fig. 9, as the contour of air temperature and wind vector on vertical plane for twelve cases (six cases for each leeward and windward emissions). As shown in Fig. 8, six test lines are located in the middle of street canyon within the target area, and quantitative analysis were conducted by averaging out the data collected from test lines. The design parameters, i.e. emission location, building height variance, building separation, building void, and coupled strategies, were evaluated to provide new understandings to support urban planners and designers to make evidence-based decision.

4.1. Emission locations

As shown in Fig. 10, the air temperature at the case with leeward emissions (black solid line) is 0.1 – 0.3 °C lower than the one with windward emissions (red solid line). It is because of the air around the heat source is heated up, and then the primary vortex

Air Temperature (°C)

Leeward position

Windward position

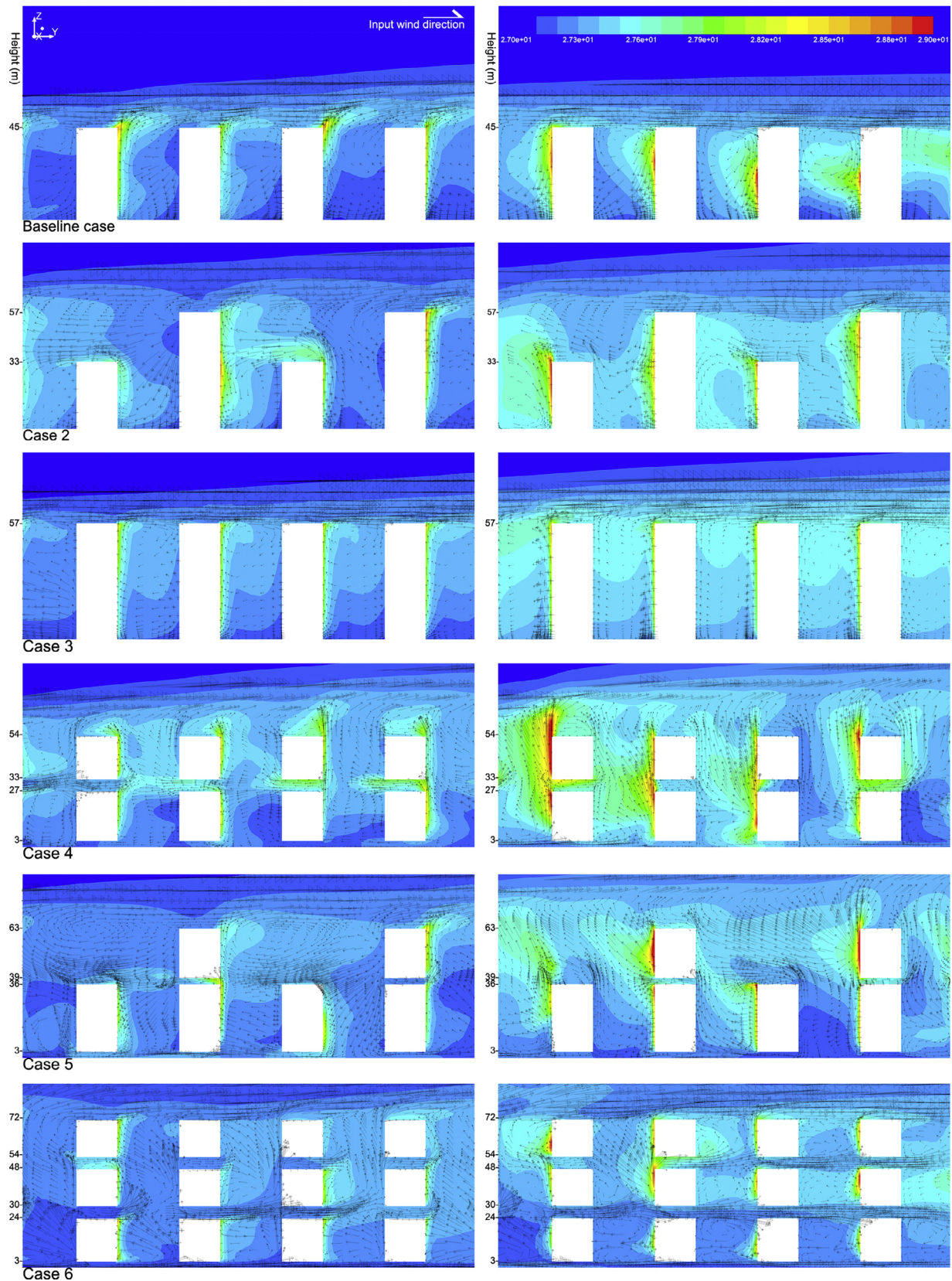


Fig. 9. Contours of air temperature and wind vector at the vertical plane of total 12 scenarios (6 cases × 2 emission locations).

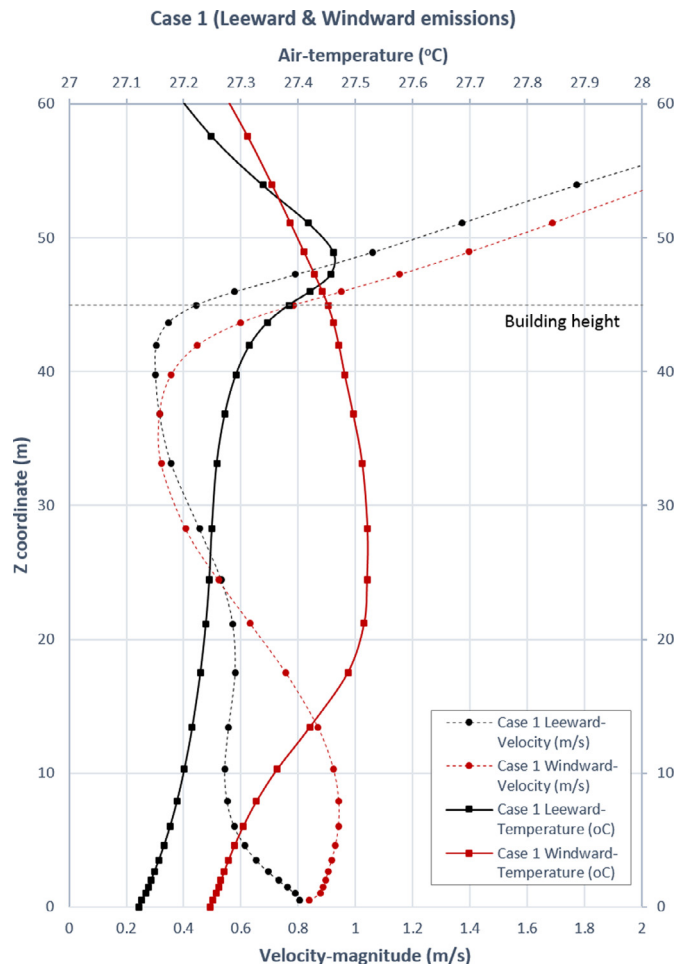


Fig. 10. Vertical profiles of average air temperature (°C) and wind velocity (m/s) at the centre of street canyons inside the test area, from Case 1 with different emission locations. Leeward emission scenario is indicated in black colour and windward emission scenario is indicated in red colour. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

in the street canyon is accelerated by the buoyancy at the leeward side. Therefore, anthropogenic heat can be removed more effectively from street canyon to the upper-layer atmosphere. Whereas for Windward emission, the downwash wind along the windward wall makes the convective heat fluxes downward into the street canyon, while the buoyancy acts in the opposite direction. Conflict between the upward buoyant force and downward force of primary vortex limits the heat removal, and causes higher air temperature

in the street canyon. Therefore, locating the heat source on leeward side is preferable when the heat is rejected from the side of the building.

It should be noted that velocity magnitude at Case 1-Windward (red dash lines) is higher than Case 1-Leeward (black dash lines), at the lower part of the street canyon, as shown in Fig. 10. It is due to the strong influence of lateral flows (Fig. 11). It has been reported by Allegrini [21] that buoyancy significantly changes the flow structure. Due to buoyancy, hot air exits in vertical direction and cold air enters from the sides of street canyon. Therefore, wind speed at the centre of the street canyon could be increased by a factor of 5–10. As suggested by Allegrini [21], this can help to ventilate the street canyon and provide cooling effect. But it should be noticed that the strong lateral flows can only be seen at the lower part of the street canyon at high density urban areas.

4.2. Building height variance

As shown in Fig. 9, the distribution of air temperature at the vertical plane is more uniform than the ones in Case 1, because building height variance increases the turbulent intensity, and thus mixing effect, in the urban canopy layer [38]. Furthermore, due to the different building heights, there are two mechanisms that influence the heat dispersion in the street canyon. In the region between higher building's rooftop level (building height 1) and lower building's rooftop level (building height 2), the primary vortex appears, which is helpful to remove the heat out from this region. While at the lower region of the street canyon build (building height 2 and below), the heated air flows over the rooftop of lower building and re-enters the street canyon as downwash along the higher building's wall. Here, air is re-entered from the opposite direction as the input wind direction and therefore the direction of secondary vortex formed in this area is reversed, from clockwise to anti-clockwise. The data analysis in Fig. 12 is consistent with the above discussion. In Fig. 12 (left), for leeward scenario, the temperature is higher than the baseline case at the region below building height 2. The heat is trapped because the downwash flows against the buoyancy along the lower half of the leeward wall of higher buildings. As for the windward scenario, the upwards motion along the windward wall of lower buildings induced by the counter-rotating secondary vortex pushes the heated air up out of the lower canyon, thus reduced the temperature as shown in Fig. 12 (right). Consequently, it is clear that building height variance should be carefully applied into the design practice to address the anthropogenic heat issue, since the forced convection promoted by building height variance could also weaken the heat removal, especially in the lower part of street canyon.

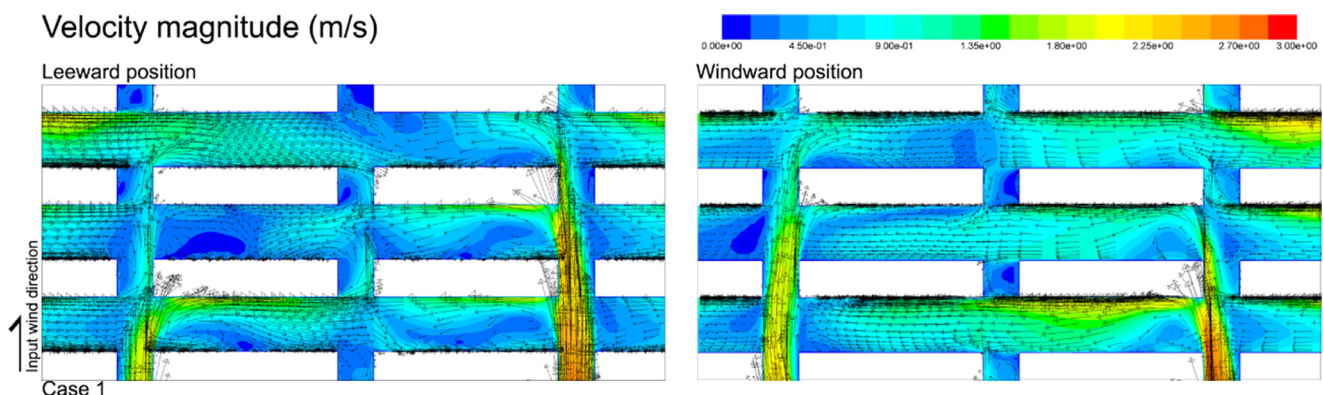


Fig. 11. Lateral flows shown at horizontal planes 10m above the ground.

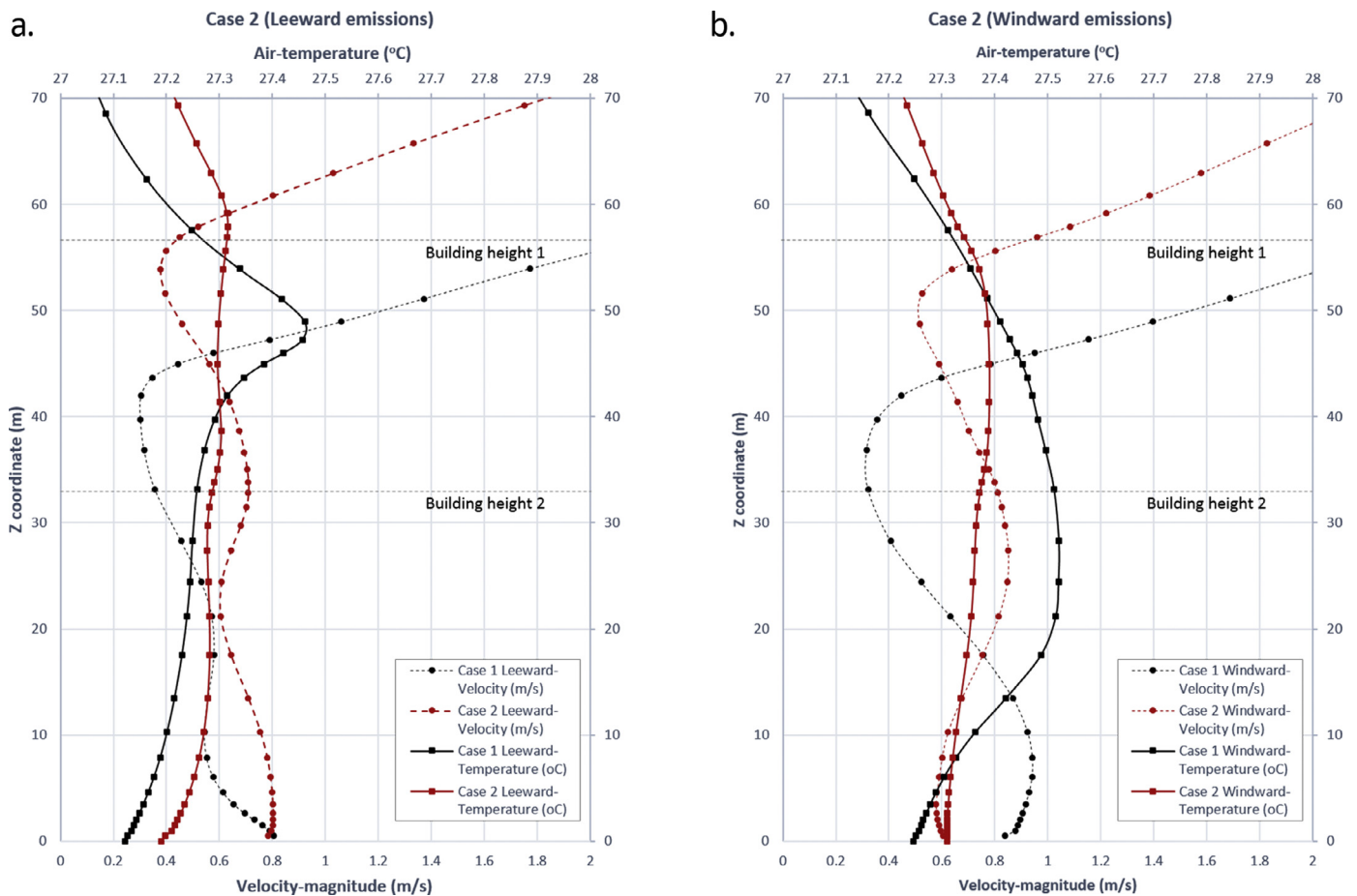


Fig. 12. Vertical profiles of average air temperature ($^{\circ}\text{C}$) and wind velocity (m/s) at the centre of street canyons inside the test area of Case 2 with different emission locations, (a) Leeward emissions and (b) Windward emissions. Baseline case, which is indicated in black colour, is included for the cross-comparison. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

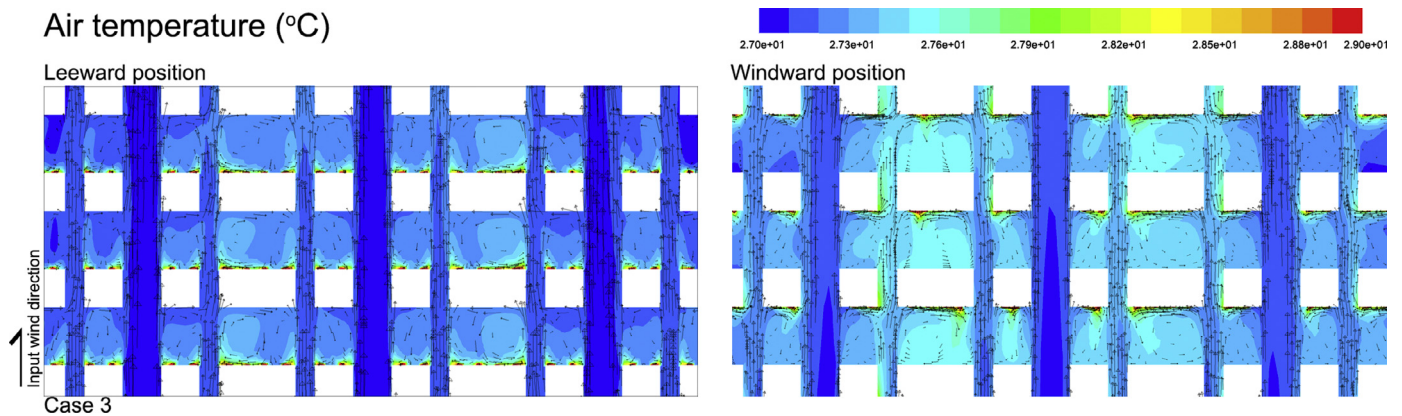


Fig. 13. Air temperature and wind vector on the horizontal plane of Case 3, at 28 m above ground ($z=0.5h$), showing the channel effect due to building gaps.

4.3. Building separation and void space

The horizontal air temperature distributions in the case with building separation (Case 3, Fig. 13) are significantly affected by the channel effect between building towers. As shown in Fig. 13, building separation can enhance anthropogenic heat removal by air flow penetration within the vertical gaps. But, on the other hand, it should be noted that anthropogenic heat could be trapped in the area between the tall towers so that the air temperature is even higher than the one at Case 1, as shown in Fig. 14 (solid lines). Therefore, at such areas, the effect of building separation on an-

thropogenic heat removal is not as positive as expected. One of the reasons is the weak vertical turbulent heat flux, due to the deeper street canyon. Another reason is the weak mixed convective heat flux. As shown in Fig. 14, a very low wind speed (less than 0.5 m/s) is observed at the vertical plane (red dash lines), which is much lower than the wind speed in Case 1 (black dash lines). This low wind speed is caused by several reasons, i.e. 1) weak lateral flow from street sides due to low vertical motion of air, and 2) clockwise primary air vortex (forced convection) against the buoyancy-driven flow (natural convection) in windward emission scenario. The above observation is different with our previous

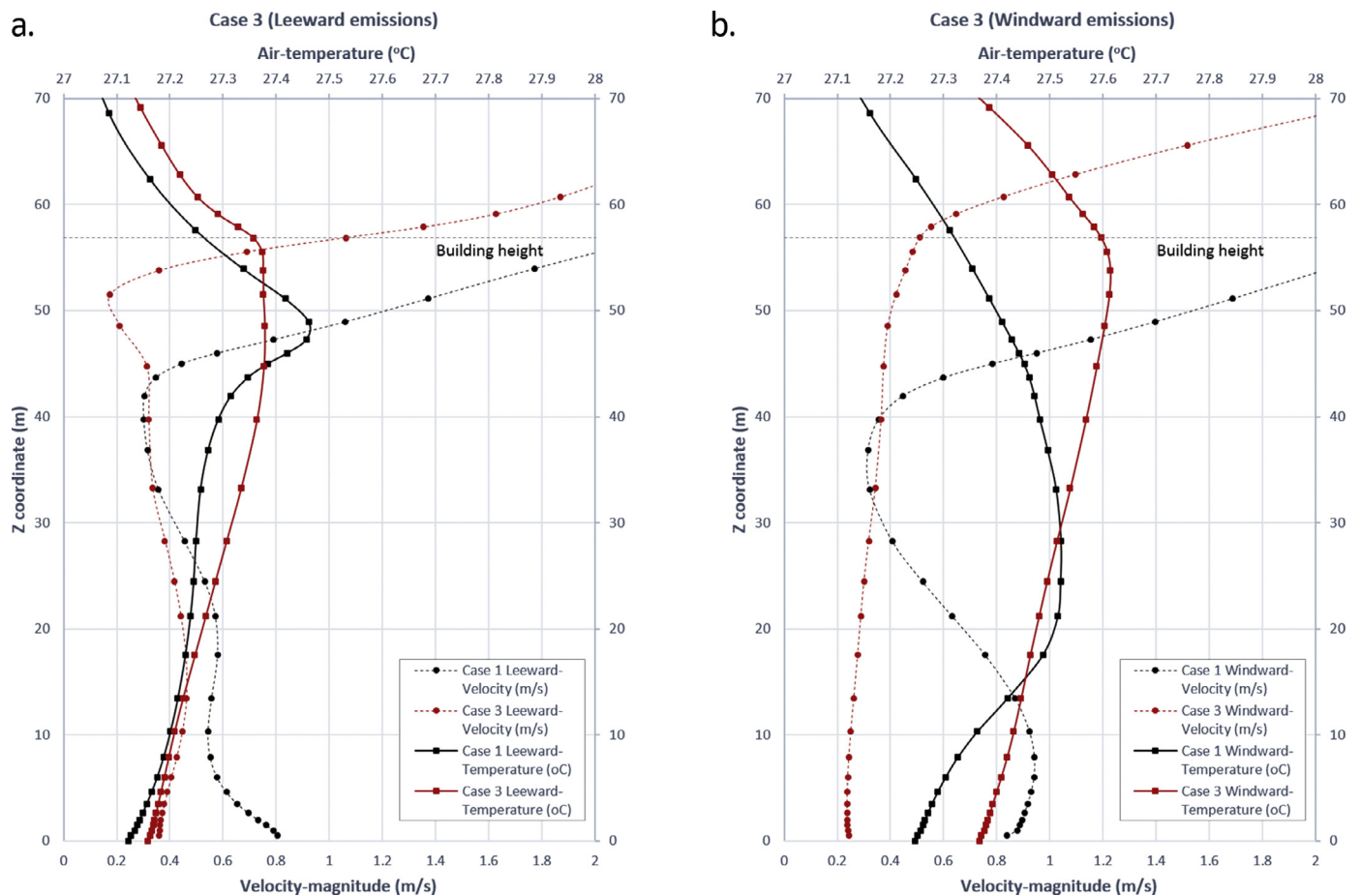


Fig. 14. Vertical profiles of average air temperature ($^{\circ}\text{C}$) and wind velocity (m/s) at the centre of street canyons inside the test area of Case 3 with different emission locations, (a) Leeward emissions and (b) Windward emissions. Baseline case, which is indicated in black colour, is included for the cross-comparison. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

studies in neutral atmosphere condition [36,37], and indicates that the building separation can significantly improve the natural ventilation and heat removal in the gaps, but, with buoyancy effect, it cannot benefit the street canyon directly between building towers.

Instead of the vertical building separation, two horizontal building void spaces at the ground floor and the middle level of street canyon are introduced in Case 4. Considerable temperature increment can be observed at the middle-level void space as shown in Fig. 9 for both leeward and windward scenarios, and there is no significant contribution of this strategy on heat removal. The voids did not create the channel effect, by which the incoming flow is supposed to go through the whole area and remove the heat to the downstream. In contrast, coupled with buoyancy effect, the building voids caused complicated flow structure in the street canyon. Different with expected, some parts of heated air from the lower level of the building is channelled backwards inside the void space due to the strong updrafts influenced by buoyancy, as shown in Fig. 9. This backwards motion caused the increment of air temperature in this area, as shown in Fig. 15. The above discussion indicates that the building void, as one of the most popular residential design strategies, could not efficiently remove the anthropogenic heat, which could be trapped in the building void instead.

4.4. Coupled design strategies

While single design strategies were applied for Cases 2, 3 and 4, and the performance of all of them was not fully efficient, coupled strategies were developed and evaluated in Cases 5 and

6. The strategy in Case 5 is the combination between Case 2 (building height variation) and Case 4 (building void), whereas the strategy in Case 6 is coupled between Case 3 (Building separation) and Case 4.

In Case 5, by adding the void spaces in the building height variance case, air temperature inside the street canyon was reduced in both Leeward and Windward emission scenarios, compared with cases with single strategies, as shown in Fig. 16. It should be noted that upward convective heat flux in windward emission scenario was still significantly affected by the downwash at the windward side of higher buildings. However, the air temperature at the lower part of the street canyon (i.e. ground level up to the building void/lower rooftop) is still lower than the temperature in Case 1 (Baseline Case).

Case 6 showed the best performance among six cases in both leeward and windward scenarios, as shown in Fig. 17. This coupled strategy made the minimum heat concentration area, and the lowest air temperature increment at near-wall as well as the centre of street canyon. In Case 6, building porosity is maximised by combining the building separation (vertical) and building void (horizontal) in every building block. It allows the formation of air path in between buildings as well as through the void spaces, which quickly transport the heated air downstream and significantly reduce the temperature inside the street canyon. The backward air flow in the building void spaces, which trapped the heat in Case 4, was avoided in Case 6. On the other hand, the negative effect of weak vertical motions due to high aspect ratio found in building separation case (Case 3) is mitigated by horizontal flows chan-

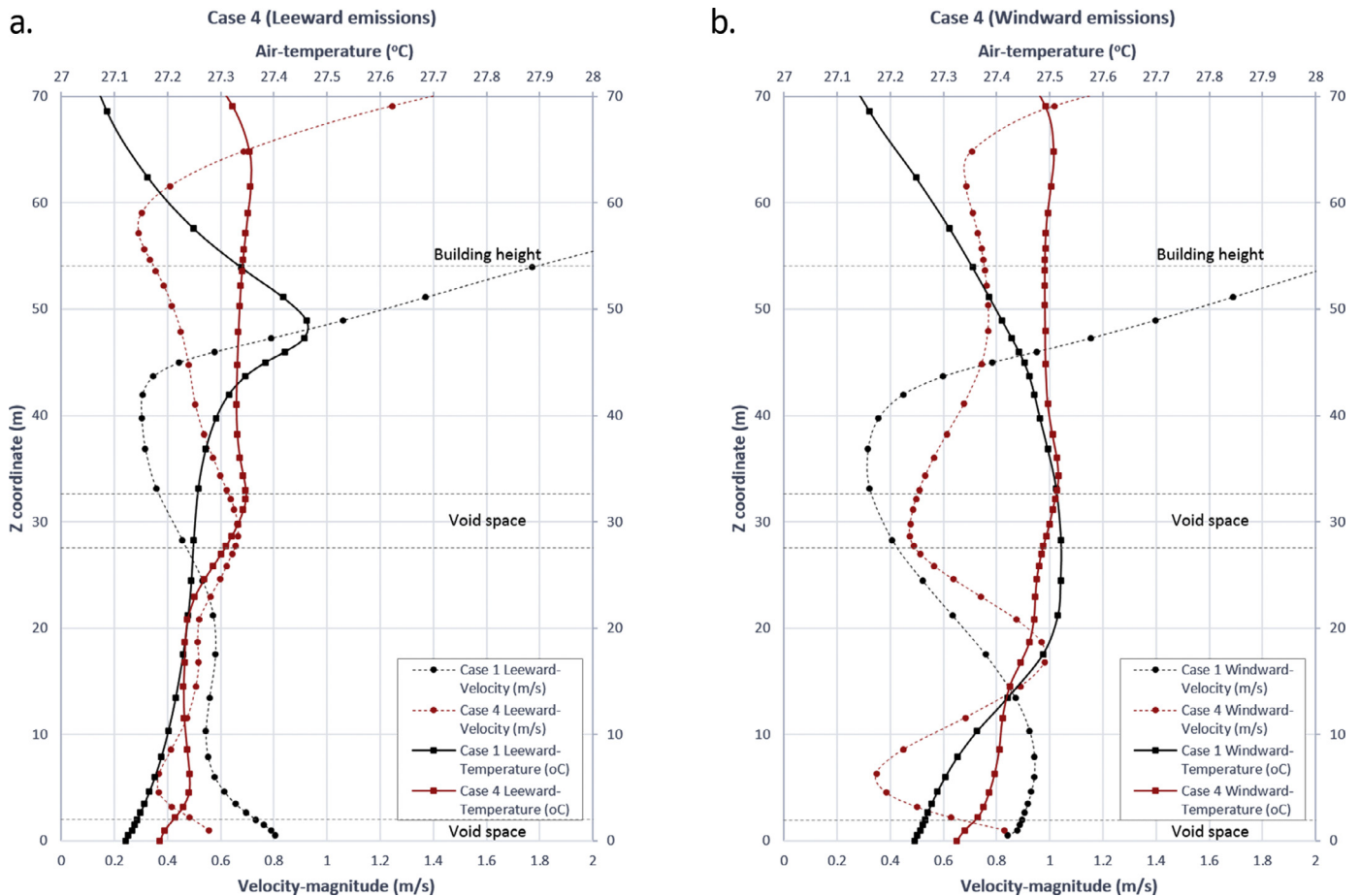


Fig. 15. Profiles of average air temperature (°C) and wind velocity (m/s) at the centre of street canyons inside the test area of Case 4 with different emission locations, (a) Leeward emissions and (b) Windward emissions. Baseline case, which is indicated in black colour, is included for the cross-comparison. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

nelled by building voids. The above analysis and discussion indicate that, compared with single strategies, coupled strategies are more reliable and efficient in removing the heat from street canyon to the upper layer atmosphere.

5. Implementation

The suggested mitigation strategies were implemented in the real urban context to see their applicability and feasibility in the real urban planning, as shown in Fig. 18. An existing Tampines new town (Case a) in Singapore was selected and redesigned based on the mitigation strategies (Case b) to improve the anthropogenic heat dispersion. For the purpose of comparison, the urban density in both cases was remained constant, while the urban permeability was increased both horizontally and vertically. Only residential blocks that are located in the centre of the precinct were modified, and the surrounding buildings are the same as the urban context. The same CFD simulation settings described previously (Section 3.2) were applied to model the anthropogenic heat dispersion, and the outputs were cross compared as shown in Fig. 19.

The existing urban morphology in the target area has uniform building height of 30 m with 1:1 aspect ratio, long slab blocks with void deck on the ground floor and AC outdoor units were placed on both sides of the street canyon. As shown in Fig. 19a, there is a large concentration area within the street canyon with temperature increase (T_{diff}) about 0.9–1.2 °C to more than 2 °C closer to the heat sources. By increasing the urban permeability as well as relocating the outdoor units to only one side of the canyon, significant

improvement can be achieved where the heat dispersion are far better than the current condition as shown in Fig. 19b. In general, the T_{diff} inside the street canyon can be reduced about 0.6 °C. This result indicates that the suggested mitigation strategies above can work effectively in the real urban planning and design.

6. Conclusion

This study aims to develop the passive cooling design strategies to mitigate the negative effect of anthropogenic heat on the thermal comfort at high density residential areas. We conducted computational parametric study, which is validated by wind tunnel experiment. Anthropogenic heat emitted from buildings was estimated based on rejected heat from air-conditioning system, including cooling load and cooling energy. In the simulation, we converted the estimated anthropogenic heat (Q_A) to the air temperature increment (ΔT) using sensible heat (Q_S) equation. Total 12 parametric cases were investigated, i.e. 1 baseline case designed based on current typology of high-density residential blocks in Singapore and 5 cases of various mitigation strategies with heat emissions either on leeward or windward side of the street canyon. The following scientific understandings on the relationship between building design and anthropogenic dispersion are provided as:

- (1) Heat rejection from leeward wall strengthen the primary vortex due to the strong updraft near the heat source. On the contrary, emissions rejected from windward side weaken the primary vortex because the reverse flow that is appeared near the

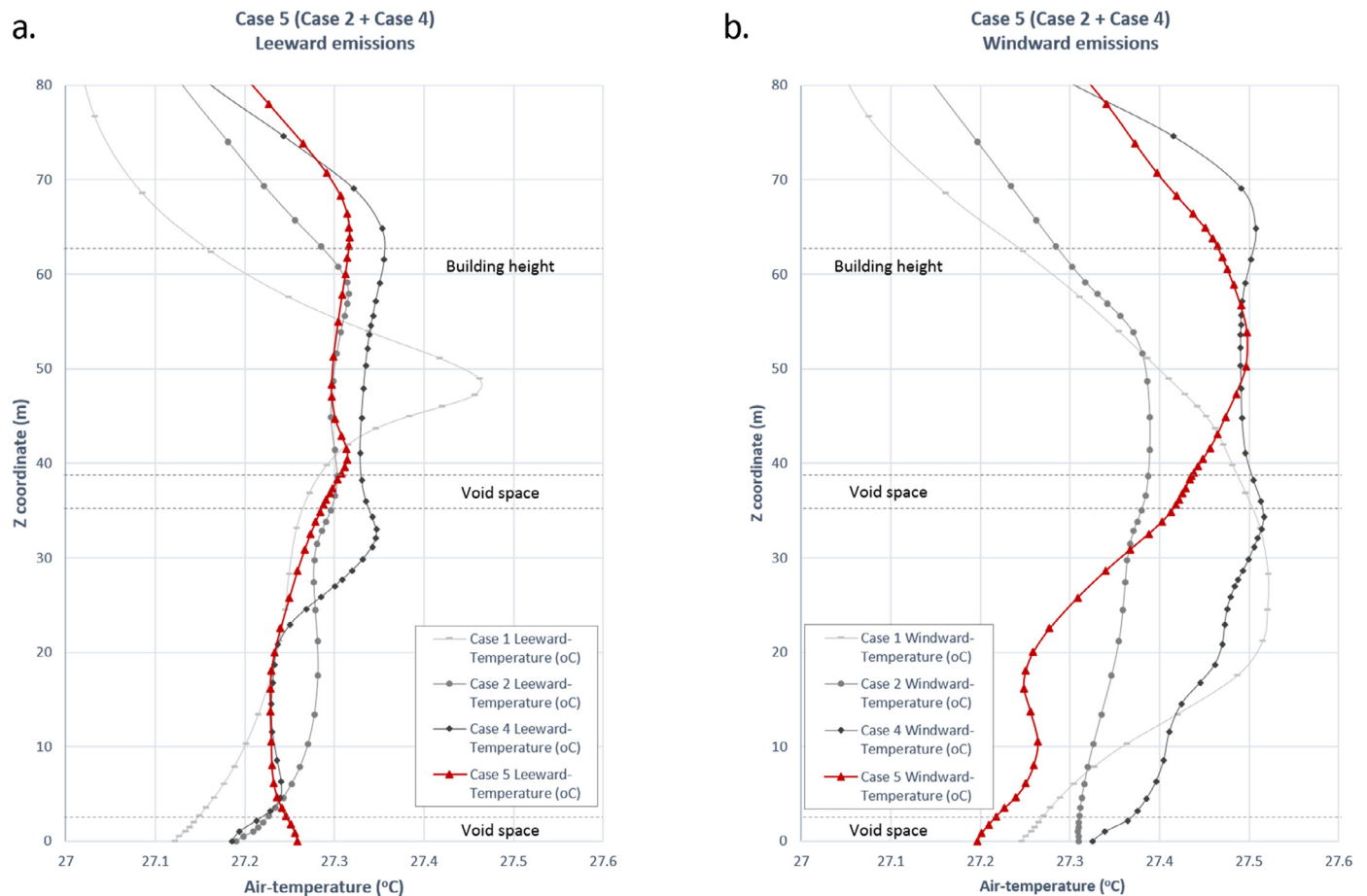


Fig. 16. Cross-comparison between Cases 1, 2, 4, 5 in terms of air temperature (°C) with different emissions' locations: (a) Leeward emissions and (b) Windward emissions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

downwind wall induced by the buoyancy. Thus, leeward position is preferable to promote a more effective vertical dispersion.

- (2) Building height variance increases the turbulent intensity thus resulting a more uniform heat distribution, and induces counter-rotating secondary vortex at the lower level of the street canyon with taller-lower building configuration. This strategy works well with windward emission, but it can trap the heat at the lower part of the street canyon and potentially increase the temperature when the heat is released from leeward position.
- (3) Building separation strategy is not as effective as expected. Vertical gaps between tall towers (i.e. point block typology) induce channel effect that can lower the temperature within the gaps. But, it should be noted that the heat can potentially be trapped inside the street canyon, in between the tall towers. The reason is that the weak turbulence and vertical motion due to the deeper street canyon weakens the induced lateral flow from the street sides thus reduces the cooling effect. Meanwhile, horizontal gaps (i.e. building voids) tends to cause a complicated flow structure in the street canyon, where some heated air from lower level is channelled backwards inside the void space due to strong updraft influenced by buoyancy. Thus, building void is not efficient in removing anthropogenic heat as it can worsen the temperature inside the street canyon as well as inside the void spaces.
- (4) Building porosity (i.e. building void and separation) is an important design aspect in promoting anthropogenic heat dispersion

that architects and urban planners need to consider. It should be noted that while adding building void alone might trap the heated air in the void space and can potentially worsen the street-canyon temperature, coupling this strategy with building separation can improve the dispersion significantly. Building separation alone has been reported to have a significant effect in enhancing urban ventilation, but in this case the impact of this strategy becomes less significant in terms of the temperature at the centre of the street canyon due to the high aspect ratio.

7. Limitation and future study

This study only considered one incoming wind direction, which is perpendicular to the street canyon as it is commonly considered as the worst case scenario in urban wind simulation. To provide more systematic building design guidelines, various incoming wind directions will be considered in the future study. Furthermore, more comprehensive design parameters will be tested, such as different aspect ratio, street canyon length, building height variance and other permutations on void space design, to provide quantitative understandings to support design decision-making. Based on the numerical simulation results from this study, we will extend the study to the city scale, and develop the GIS-based semi-empirical urban canopy model to estimate the impact of anthropogenic heat on air temperature.

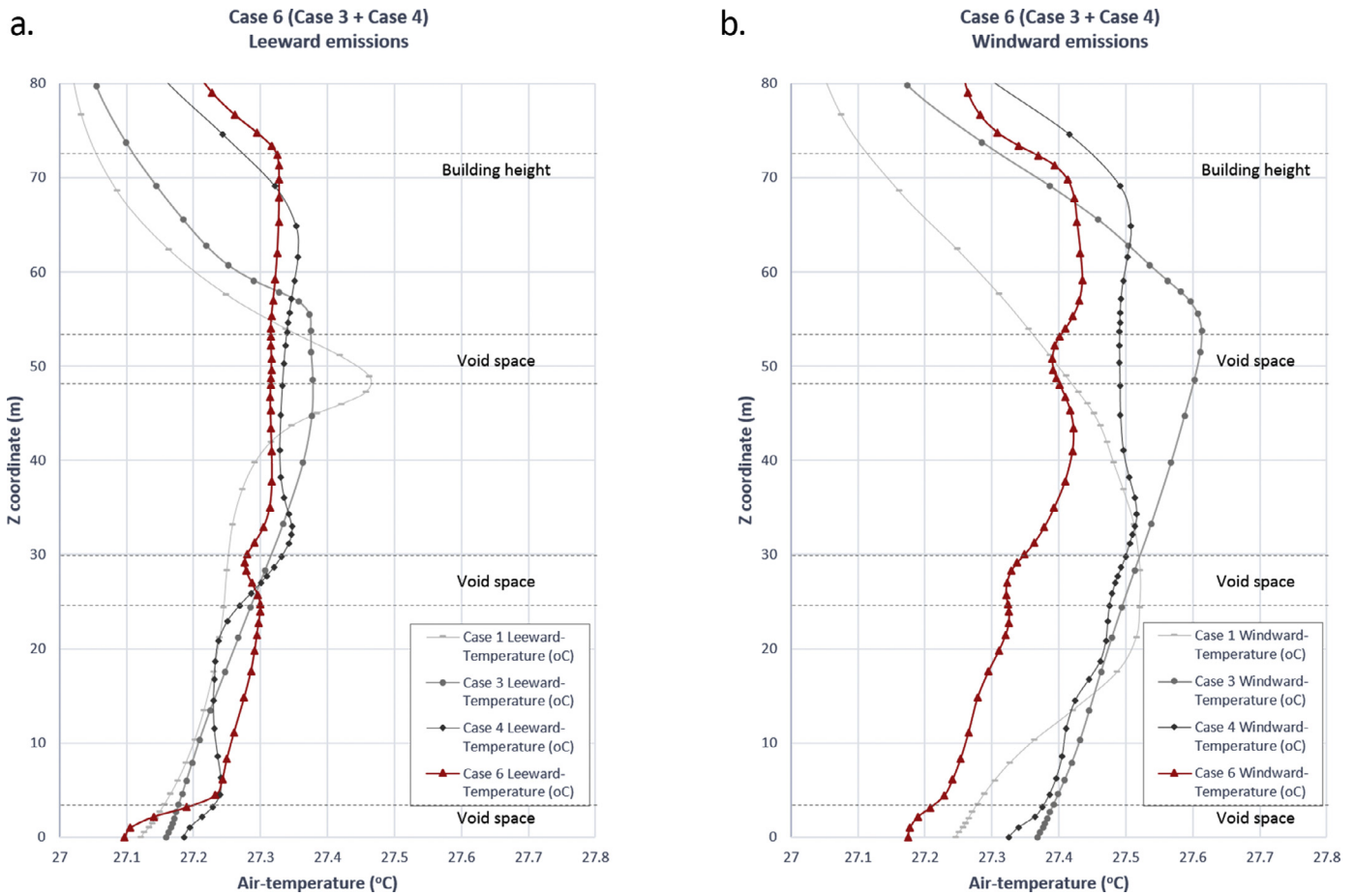


Fig. 17. Cross-comparison between Cases 1 (baseline case), 3, 4 and 6 in terms of air temperature (°C) with different emissions' locations: (a) Leeward emissions and (b) Windward emissions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

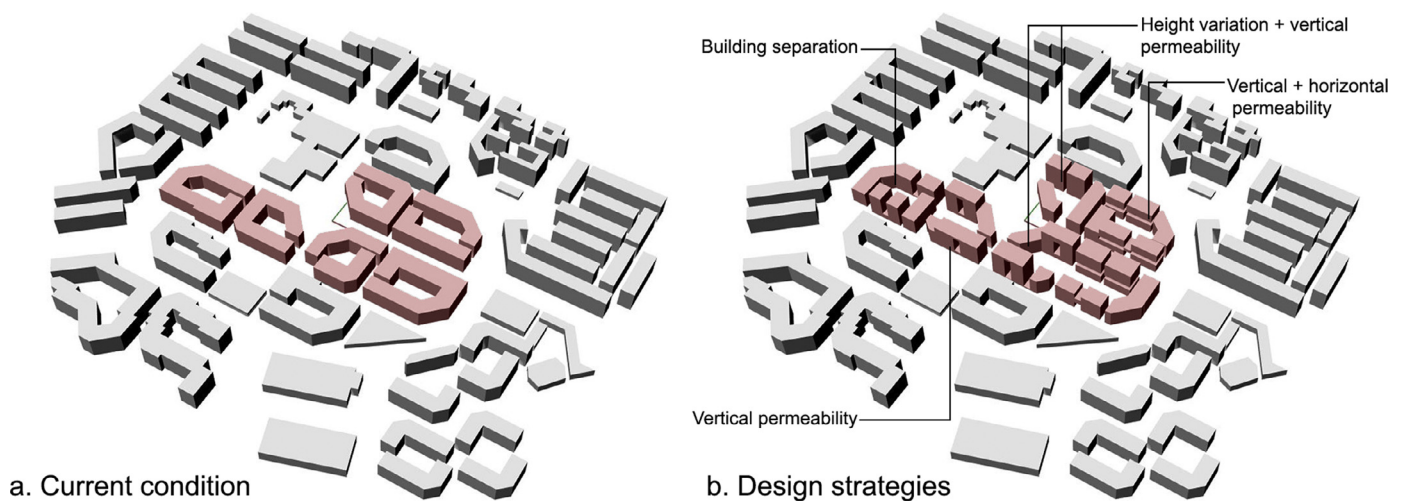


Fig. 18. Implementation of mitigation strategies in real urban planning, taking Tampines new town as case study.

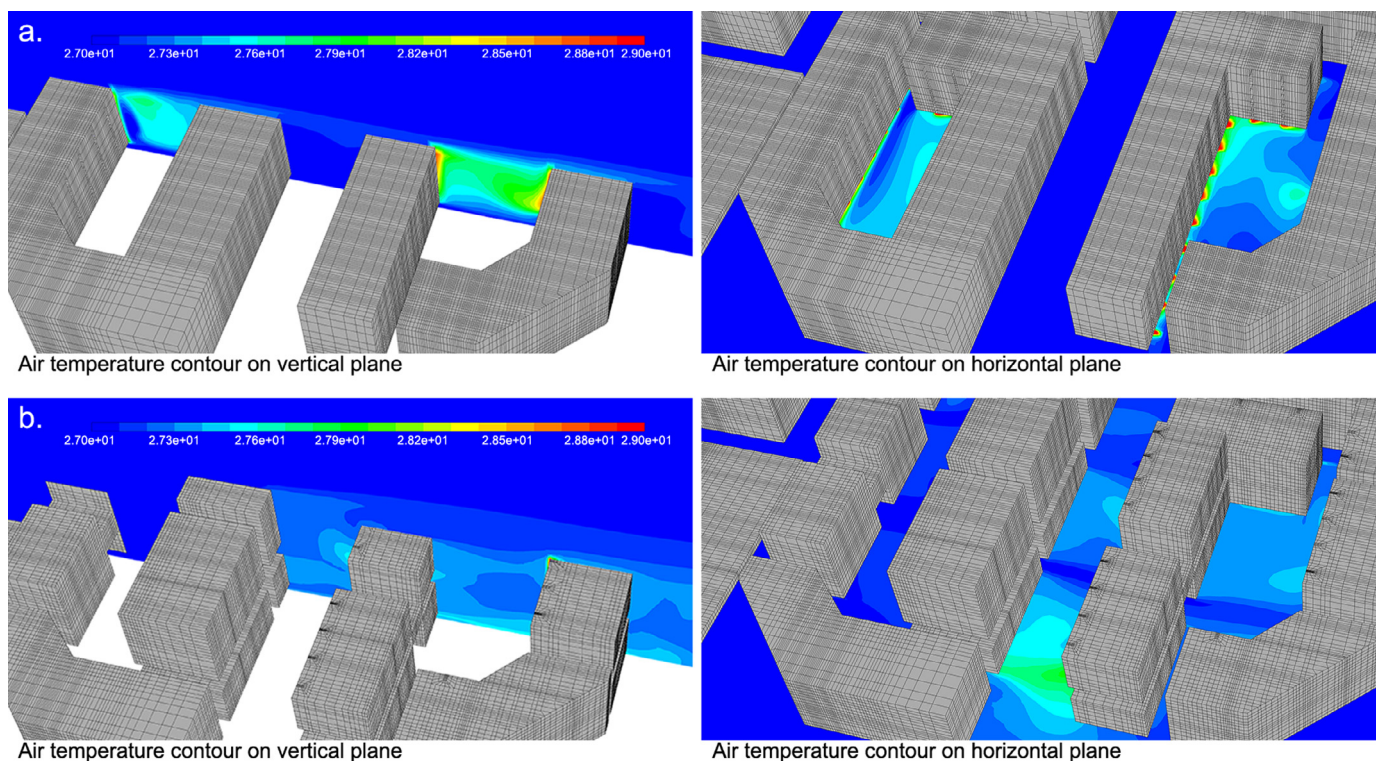


Fig. 19. Air temperature distributions at vertical and horizontal planes at Case (a) current design and Case (b) design with mitigation strategies.

Conflict of interest

There is no conflict of interest.

Acknowledgement

The authors gratefully acknowledge the financial support by the Academic Research Fund (AcRF) grant from the Ministry of Education (MOE) and the National University of Singapore (R-295-000-139-133).

References

- [1] L. Allen, F. Lindberg, C.S.B. Grimmond, Global to city scale urban anthropogenic heat flux: model and variability, *Int. J. Climatol.* 31 (13) (2011) 1990–2005.
- [2] Energy Information Administration (EIA), Annual Energy Review 2011, Government Printing Office, 2012.
- [3] C.-M. Hsieh, T. Aramaki, K. Hanaki, Managing heat rejected from air conditioning systems to save energy and improve the microclimates of residential buildings, *Comp. Environ. Urban Syst.* 35 (5) (2011) 358–367.
- [4] S. Dhakal, K. Hanaki, A. Hiramatsu, Estimation of heat discharges by residential buildings in Tokyo, *Energy Convers. Manage.* 44 (9) (2003) 1487–1499.
- [5] T.R. Oke, The energetic basis of the urban heat island, *Q. J. R. Meteorol. Soc.* 108 (455) (1982) 1–24.
- [6] M. Roth, W.T. Chow, A historical review and assessment of urban heat island research in Singapore, *Singapore J. Trop. Geog.* 33 (3) (2012) 381–397.
- [7] F. Kimura, S. Takahashi, The effects of land-use and anthropogenic heating on the surface temperature in the Tokyo Metropolitan area: a numerical experiment, *Atmos. Environ. Part B Urban Atmos.* 25 (2) (1991) 155–164.
- [8] T. Ichinose, K. Shimodono, K. Hanaki, Impact of anthropogenic heat on urban climate in Tokyo, *Atmos. Environ.* 33 (24–25) (1999) 3897–3909.
- [9] C.-M. Hsieh, T. Aramaki, K. Hanaki, Estimation of heat rejection based on the air conditioner use time and its mitigation from buildings in Taipei City, *Build. Environ.* 42 (9) (2007) 3125–3137.
- [10] Y. Ohashi, Y. Genchi, H. Kondo, Y. Kikigawa, H. Yoshikado, Y. Hirano, Influence of air-conditioning waste heat on air temperature in Tokyo during summer: numerical experiments using an urban canopy model coupled with a building energy model, *J. Appl. Meteorol. Climatol.* 46 (1) (2007) 66–81.
- [11] R. Priyadarsini, W.N. Hien, C.K. Wai David, Microclimatic modeling of the urban thermal environment of Singapore to mitigate urban heat island, *Sol. Energy* 82 (8) (2008) 727–745.
- [12] M. Bojic, M. Lee, F. Yik, Influence of a depth of a recessed space to flow due to air-conditioner heat rejection, *Energy Build.* 34 (1) (2002) 33–43.
- [13] Y. Ashie, T. Kono, Urban-scale CFD analysis in support of a climate-sensitive design for the Tokyo Bay area, *Int. J. Climatol.* 31 (2) (2011) 174–188.
- [14] A. Dallman, S. Magnusson, R. Britter, L. Norford, D. Entekhabi, H.J.S. Fernando, Conditions for thermal circulation in urban street canyons, *Build. Environ.* 80 (2014) 184–191.
- [15] K. Uehara, S. Murakami, S. Oikawa, S. Wakamatsu, Wind tunnel experiments on how thermal stratification affects flow in and above urban street canyons, *Atmos. Environ.* 34 (10) (2000) 1553–1562.
- [16] A. Kovar-Panskus, L. Moulinneuf, E. Savory, A. Abdelqari, J.-F. Sini, J.-M. Rosant, N. Toy, A wind tunnel investigation of the influence of solar-induced wall-heating on the flow regime within a simulated urban street canyon, *Water Air Soil Pollut. Focus* 2 (5) (2002) 555–571.
- [17] J. Allegrini, V. Dorer, J. Carmeliet, Wind tunnel measurements of buoyant flows in street canyons, *Build. Environ.* 59 (Supplement C) (2013) 315–326.
- [18] J.-J. Kim, J.-J. Baik, A numerical study of thermal effects on flow and pollutant dispersion in urban street canyons, *J. Appl. Meteorol.* 38 (9) (1999) 1249–1261.
- [19] S.-B. Park, J.-J. Baik, S. Raasch, M.O. Letzel, A large-eddy simulation study of thermal effects on turbulent flow and dispersion in and above a street canyon, *J. Appl. Meteorol. Climatol.* 51 (5) (2012) 829–841.
- [20] X. Xie, C.-H. Liu, D.Y.C. Leung, Impact of building facades and ground heating on wind flow and pollutant transport in street canyons, *Atmos. Environ.* 41 (39) (2007) 9030–9049.
- [21] J. Allegrini, A wind tunnel study on three-dimensional buoyant flows in street canyons with different roof shapes and building lengths, *Build. Environ.* 143 (2018) 71–88.
- [22] J.-J. Kim, J.-J. Baik, Effects of street-bottom and building-roof heating on flow in three-dimensional street canyons, *Adv. Atmos. Sci.* 27 (3) (2010) 513–527.
- [23] Singstat, Population Trends 2016, Department of Statistics, ministry of trade & Industry, Republic of Singapore, Singapore, 2016 Retrieved from http://www.singstat.gov.sg/docs/default-source/default-document-library/publications/publications_and_papers/population_and_population_structure/population2016.pdf.
- [24] M. Bojic, M. Lee, F. Yik, Flow and temperatures outside a high-rise residential building due to heat rejection by its air-conditioners, *Energy Build.* 33 (7) (2001) 737–751.
- [25] Fluent 12. User guide. Fluent Inc. Retrieved from: http://www.afs.enea.it/project/neptunius/docs/fluent/html/ug/main_pre.htm
- [26] Y. Tominaga, A. Mochida, R. Yoshie, H. Kataoka, T. Nozu, M. Yoshikawa, T. Shirasawa, AIJ guidelines for practical applications of CFD to pedestrian wind environment around buildings, *J. Wind Eng. Indust. Aerodynam.* 96 (10) (2008) 1749–1761.
- [27] Y. Kikigawa, Y. Genchi, H. Yoshikado, H. Kondo, Development of a numerical simulation system toward comprehensive assessments of urban warming countermeasures including their impacts upon the urban buildings' energy-demands, *Appl. Energy* 76 (4) (2003) 449–466.

- [28] S.L. Fun, Public Housing Design Handbook, Architectural Department, Housing & Development Board (HDB), Singapore, 1995.
- [29] HDB, Key Statistics, Annual Report 2016/2017, 2017. Retrieved from <http://www10.hdb.gov.sg/ebook/AR2017/key-statistics.html>.
- [30] National Environment Agency (NEA), Energy label and Tick Rating prior to 1 September 2014, 2014. Retrieved from <https://www.nea.gov.sg>.
- [31] Quah, A.K.L., & Roth, M. (2012). Diurnal and weekly variation of anthropogenic heat emissions in a tropical city, Singapore.
- [32] M. Bruehlisauer, F. Meggers, E. Saber, C. Li, H. Leibundgut, Stuck in a stack—Temperature measurements of the microclimate around split type condensing units in a high rise building in Singapore, *Energy Build.* 71 (Supplement C) (2014) 28–37.
- [33] T.S. Foo, Planning and design of Tampines, an award-winning high-rise, high-density township in Singapore, *Cities* 18 (1) (2001) 33–42.
- [34] R.X. Lee, S.K. Jusuf, N.H. Wong, The study of height variation on outdoor ventilation for Singapore's high-rise residential housing estates, *Int. J. Low-Carbon Technol.* 10 (1) (2015) 15–33.
- [35] W.T.L. Chow, M. Roth, Temporal dynamics of the urban heat island of Singapore, *Int. J. Climatol.* 26 (15) (2006) 2243–2260.
- [36] C. Yuan, E. Ng, Building porosity for better urban ventilation in high-density cities – A computational parametric study, *Build. Environ.* 50 (2012) 176–189.
- [37] C. Yuan, E. Ng, L.K. Norford, Improving air quality in high-density cities by understanding the relationship between air pollutant dispersion and urban morphologies, *Build. Environ.* 71 (2014) 245–258.
- [38] J. Hang, Q. Wang, X. Chen, M. Sandberg, W. Zhu, R. Buccolieri, S. Di Sabatino, City breathability in medium density urban-like geometries evaluated through the pollutant transport rate and the net escape velocity, *Build. Environ.* 94 (2015) 166–182.